

REVIEW

**HIGHLIGHTS OF CEREBROVASCULAR DISEASES
AND
CENTRAL NERVOUS SYSTEM TRAUMA**

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DEFINITIONS

Tissue based definitions according to consensus statements from the American Heart Association and American Stroke Association (AHA/ASA)

TRANSIENT ISCHEMIC ATTACK (TIA):

Transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia, **without** acute infarction.

CENTRAL NERVOUS SYSTEM (CNS) INFARCTION:

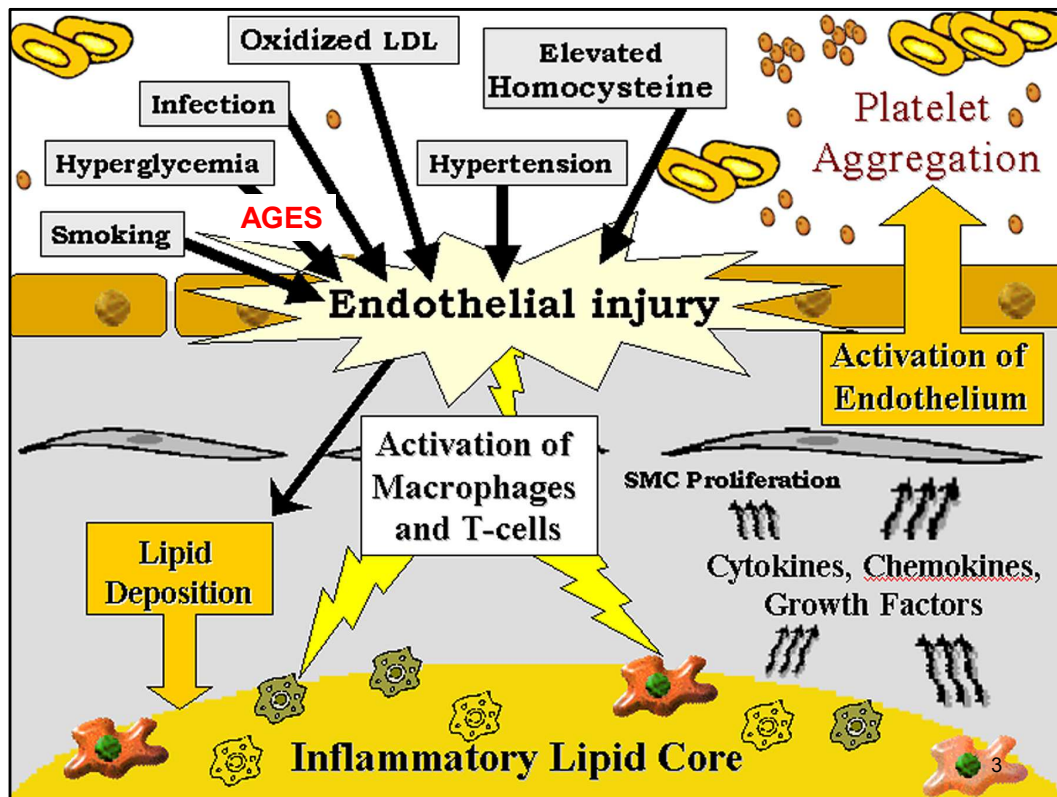
Brain, spinal cord, or retinal cell death attributable to ischemia, based on neuropathologic, neuroimaging, and/or clinical evidence of permanent injury.

ISCHEMIC STROKE:

An episode of neurologic dysfunction caused by focal cerebral, spinal, or retinal infarction.

SILENT CNS INFARCTION:

Imaging or neuropathological evidence of CNS infarction without a history of acute neurological dysfunction



Genetic and inflammatory mechanisms in stroke. Gray background = subendothelium; white background = blood vessel lumen; SMC = intraluminal smooth muscle cells.

CAUSES OF ISCHEMIC INFARCTION

- Atherosclerosis-atherothrombosis
- Small vessel disease
- Cardioembolism, including paradoxical embolism through a patent foramen ovale
- Vasculitis
- Arterial dissection
- Polycythemia, thrombocythemia, thrombotic thrombocytopenic purpura (TTP)
- Systemic lupus erythematosus (SLE)
- Sickle cell disease
- Vascular spasm
- Meningitis and fungal vasculitis
- Hypercoagulability (Factor V Leiden, Prothrombin 20210A mutation, antiphospholipid antibody syndrome)
- Inherited metabolic disorders (Fabry disease, homocystinuria, mitochondrial disorders)
- Fibromuscular dysplasia and other angiopathies

INFARCT

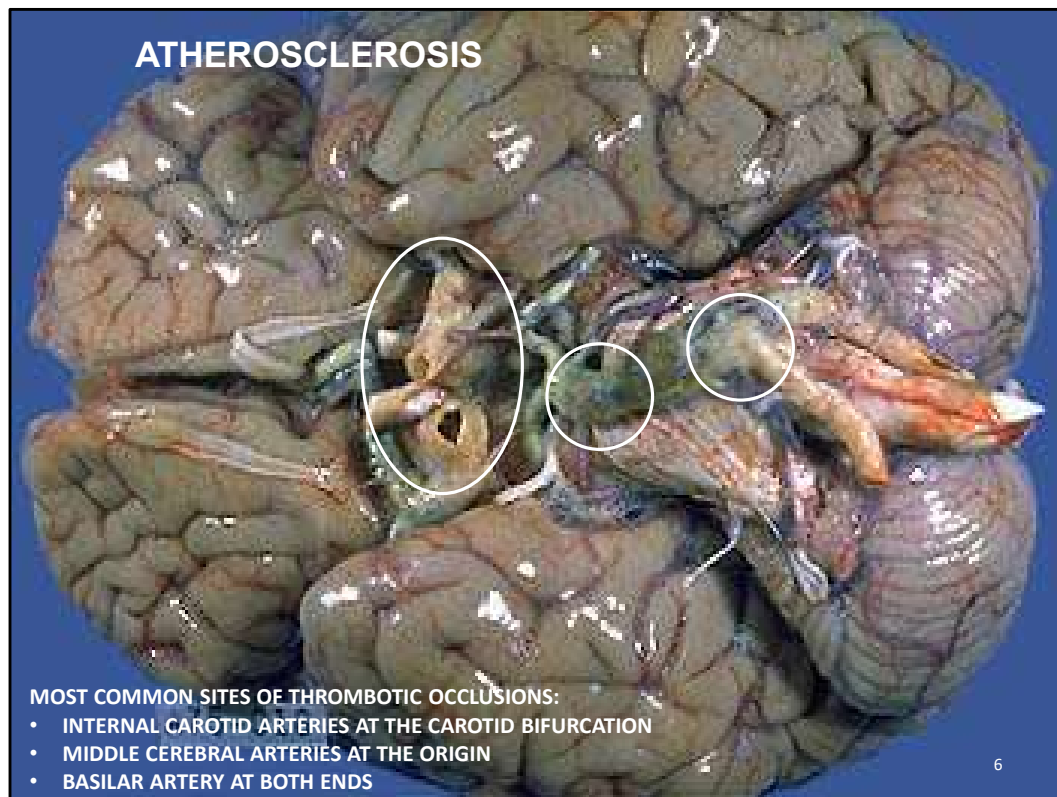
THROMBOTIC OCCLUSIONS:

Most common sites:

- Internal carotid arteries at the carotid bifurcation
- Middle cerebral arteries at the origin
- Basilar artery at both ends

EMBOLIC OCCLUSIONS:

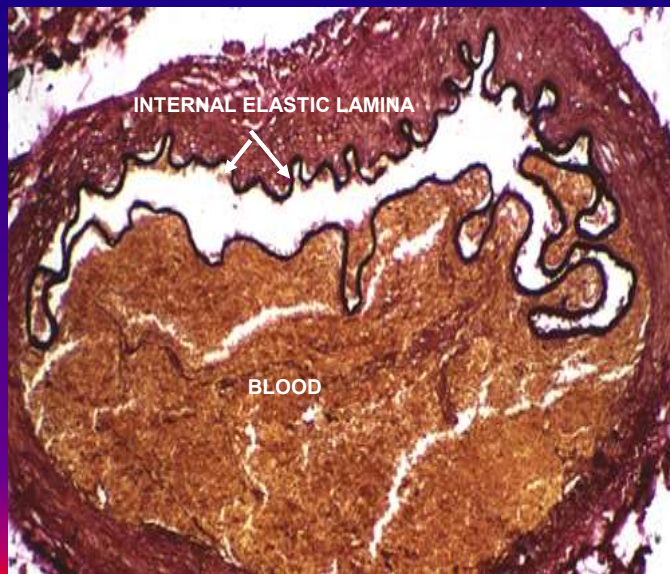
- Most common cause of an intracranial arterial occlusion
- Thromboemboli and emboli of atheromatous material most commonly originate in:
 - Heart
 - Carotids
 - More proximal arterial segment
- Less common emboli:
 - Vegetations from endocarditis
 - Fat
 - Bone marrow
 - Air



Brain. Severe atherosclerosis in internal carotid arteries, basilar and vertebral arteries and in the circle of Willis.

INFARCT

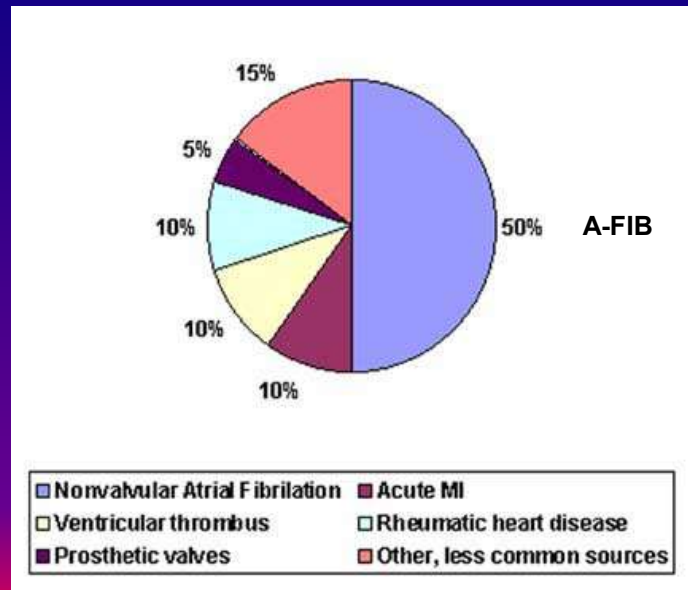
DISSECTING ANEURYSM IN CAROTID ARTERY



- DISSECTION OF THE CAROTID AND VERTEBRAL ARTERIES CAUSES ARTERIAL OCCLUSION AND EMBOLISM.
- ACCOUNTS FOR UP TO 20% OF STROKES IN YOUNGER PATIENT.

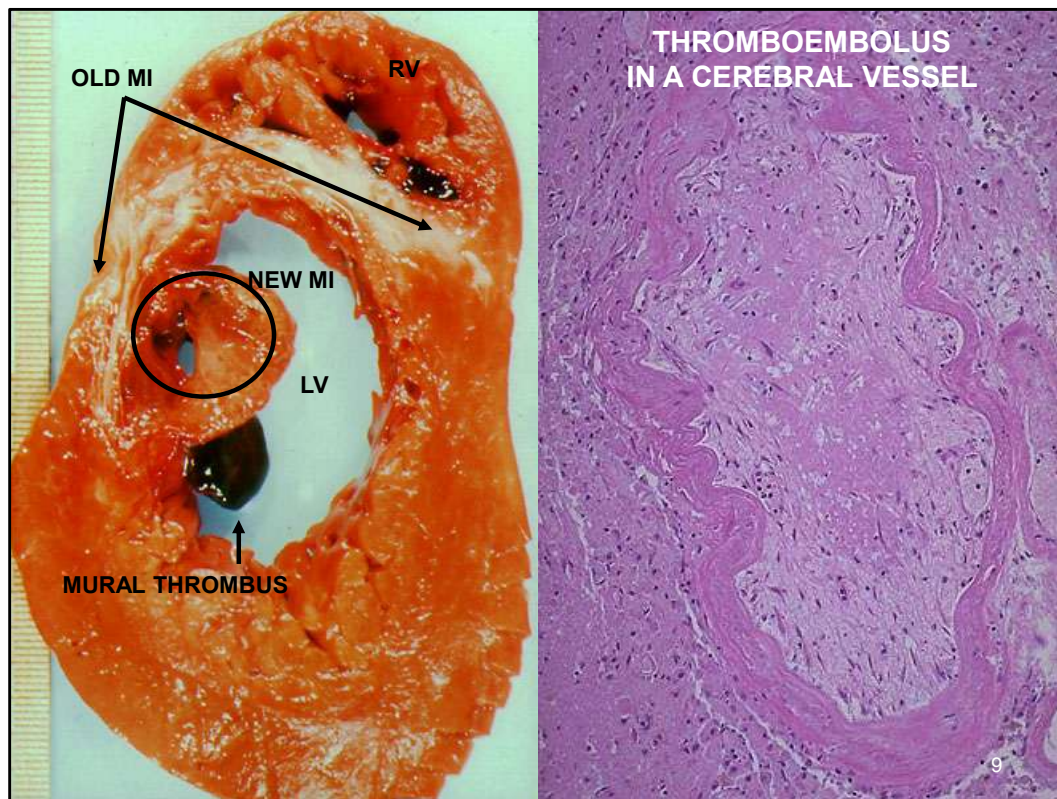
Dissecting aneurysm in carotid artery of a young man. Blood has dissected between the internal elastic lamina and the media and is obstructing blood flow to the brain. Elastic van Gieson stain

SOURCES OF CARDIOEMBOLIC STROKE

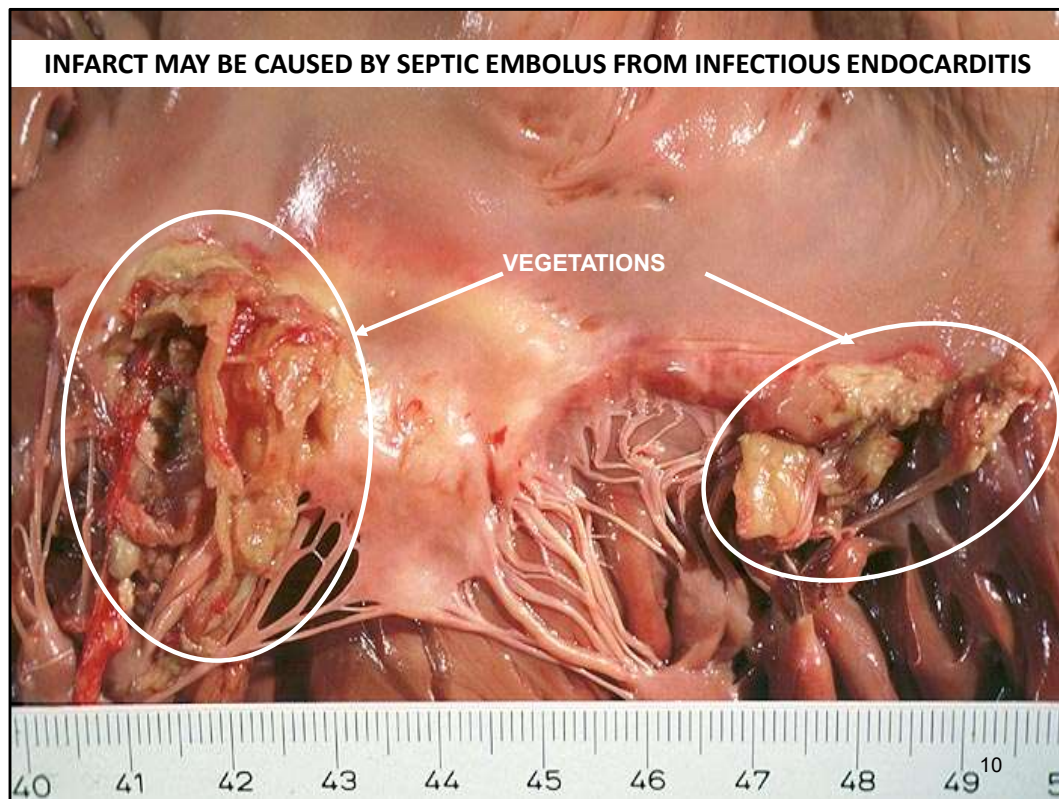


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Sources of cardioembolic stroke.

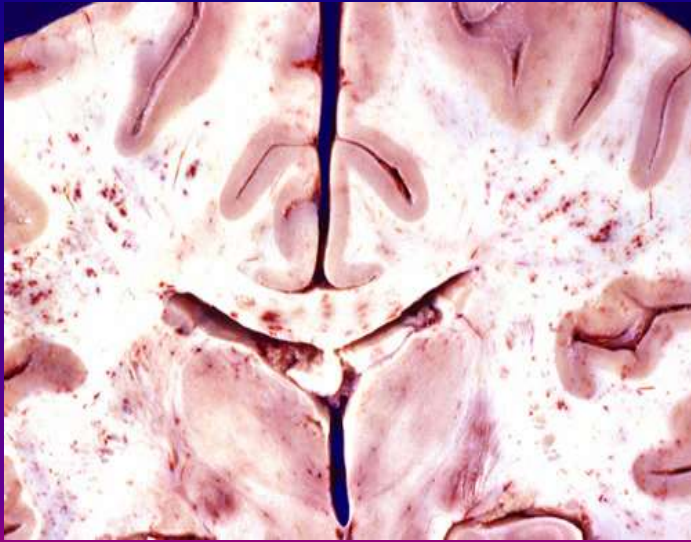


Left: Complication of MI: mural thrombus attached to damaged ventricular wall. Old infarct appears white. **Right:** Here is a thromboembolus that originated from mural thrombus in the left atrium. The heart is a common source for such emboli.



Diagnosis: infectious endocarditis with septic emboli to the brain.

FAT EMBOLISM SYNDROME



EXTENSIVE WHITE MATTER PETECHIAL HEMORRHAGES

COMPLICATION OF LONG BONE FRACTURES AND TRAUMATIC LESIONS OF VISCERA AND SOFT TISSUES.

40-60% OF PATIENTS WITH TOTAL JOINT ARTHROPLASTY HAVE FAT EMBOLISM, BUT ONLY 1-5% DEVELOP THE SYNDROME.

NEUROLOGIC SIGNS AND SYMPTOMS TYPICALLY PRESENT 24-72 HRS. AFTER THE INITIAL TRAUMA, BUT MAY START UP TO A WEEK LATER.

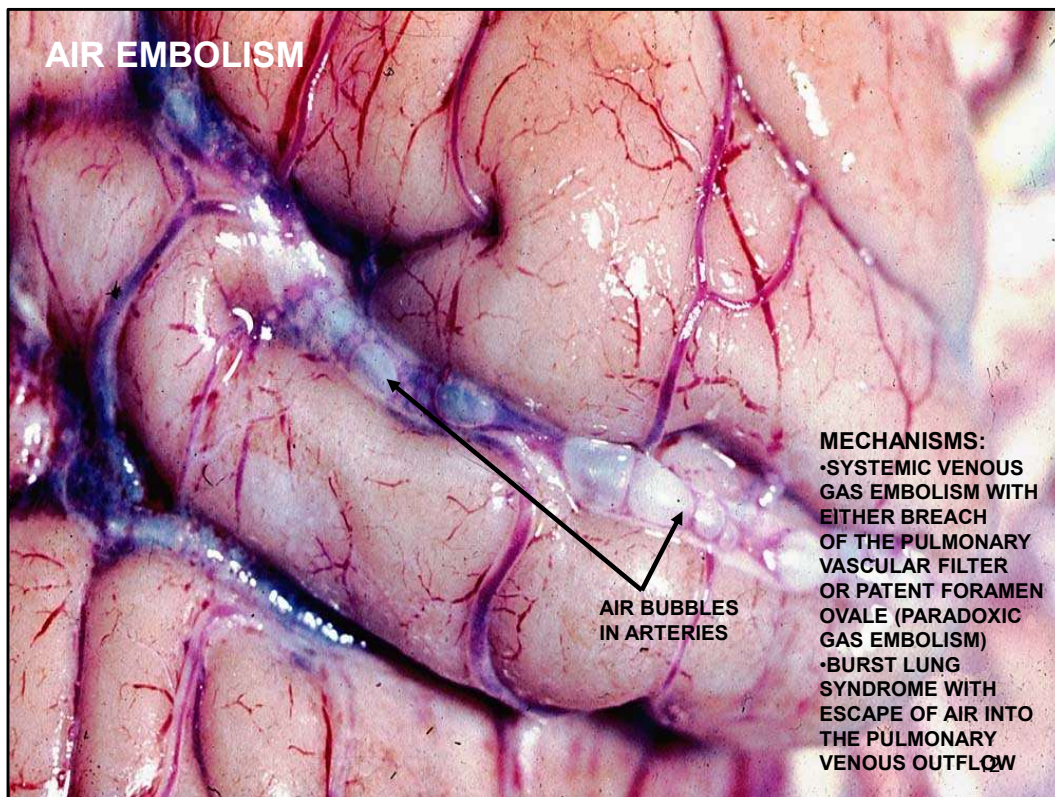
FAT GLOBULES CAN BE DEMONSTRATED IN LUNG POST-CAPILLARY VENULES INDICATING THAT ASD OR VSD (ALLOWING PARADOXIC EMBOLIZATION) ARE NOT NECESSARY.

NON-TRAUMATIC SETTINGS:

- OSTEOMYELITIS
- PANCREATITIS
- DIABETES

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The extensive white matter petechial hemorrhages seen here are typical for fat embolism syndrome. The patient manifests generalized cerebral dysfunction with disturbances of higher cortical function and consciousness, often without localizing signs, usually about a week after the initiating event, such as long bone fractures in a vehicular accident.



Brain. Diagnosis: Air embolism. **Description:** Numerous intraluminal air bubbles are visible. **Comment:** Bubbles in the arterial circulation can arise from basically three sources: venous gas embolism with breach of the pulmonary vascular filter (paradoxical gas embolism), patent foramen ovale (paradoxical gas embolism) and tear of the pulmonary parenchyma with entry of gas into pulmonary venous outflow. Studies show that systemic venous bubbles are trapped in the pulmonary arterial tree and are usually completely eliminated from that site. The other main mechanism for arterial gas embolism is by way of the pulmonary overpressure syndrome or 'burst lung'. This occurs from baropressure increases as the diver on compressed air ascends with a closed glottis or a free diver takes a breath of compressed air at depth and ascends. Because of Boyle's law, maximal changes in volume occur in the 4 feet (1.22 m.) closest to the surface and the diver sustains a tear in the pulmonary parenchyma with the escape of air into the pulmonary venous outflow.

LESS COMMON CAUSES OF INFARCTS

INFECTIOUS VASCULITIS

- Syphilis and TB used to be common causes
- Now more common in the setting of immunosuppression and opportunistic infection
 - Aspergillosis and mucor most common
 - CMV encephalitis

COLLAGEN VASCULAR DISEASE

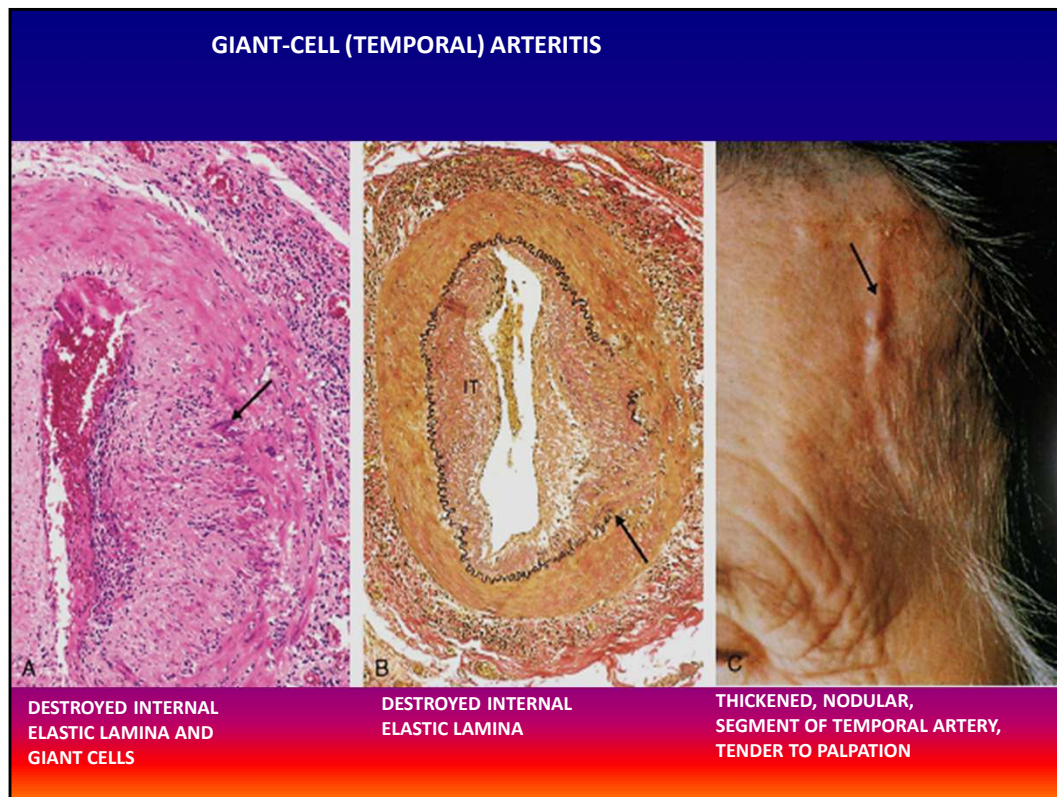
- SLE, RA, Bechet

NON-INFECTIOUS SYSTEMIC VASCULITIS WITH CNS INVOLVEMENT

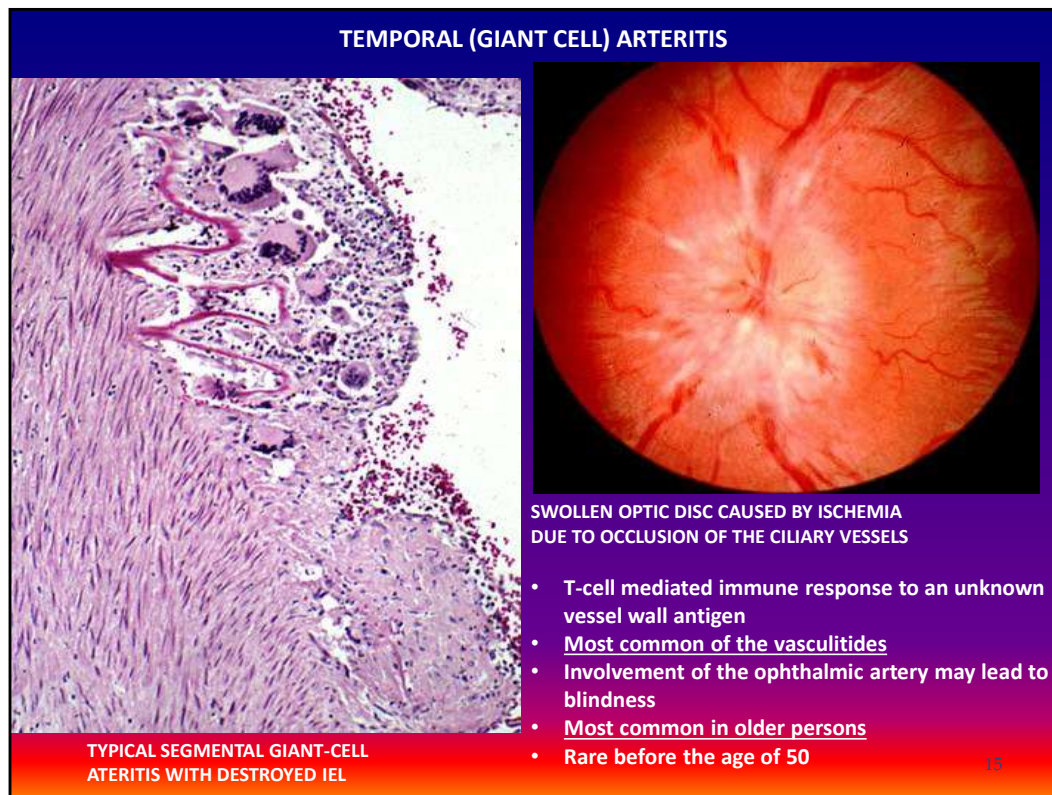
- Giant cell (temporal) arteritis most common
- Polyarteritis nodosa, Wegener granulomatosis, Takayasu arteritis, Kawasaki disease

PRIMARY GRANULOMATOUS ANGIITIS OF THE CNS

- Involves multiple small- to medium-sized parenchymal and subarachnoid vessels and causes destruction of the vessel walls
- Clinically diffuse encephalopathy or multifocal manifestations and cognitive dysfunction
- Patients improve with steroids and immunosuppressive therapy



Robbins and Cotran Pathologic Basis of Disease, Professional Edition , 10th Ed.
FIGURE 11-23 Giant-cell (temporal) arteritis. **A**, H&E stain of section of temporal artery showing giant cells at the degenerated internal elastic lamina in active arteritis (*arrow*). **B**, Elastic tissue stain demonstrating focal destruction of internal elastic lamina (*arrow*) and intimal thickening (IT) characteristic of long-standing or healed arteritis. **C**, Examination of the temporal artery of a patient with giant-cell arteritis shows a thickened, nodular, and tender segment of a vessel on the surface of head (*arrow*). (*C* from Salvarani C et al.: Polymyalgia rheumatica and giant-cell arteritis. *N Engl J Med* 347:261, 2002.)



Left: Giant cell arteritis. **Right:** Swollen optic disc caused by ischemia due to occlusion of the ciliary feeder vessels of the optic nerve. The patient experiences sudden, painless, usually devastating, permanent loss of vision in one eye. Most individuals are aged over 70 years and have a combination of limb girdle myalgias, fever and lassitude ("polymyalgia rheumatica"), caused by an idiopathic autoimmune disorder of the elderly. Some patients have scalp tenderness and pain with chewing, which may be manifestations of external carotid artery ischemia.

Laboratory findings: elevated sedimentation rate and C-reactive protein. Biopsy of a temporal artery shows granulomatous inflammation.

The diagnosis must be made immediately because there is a strong chance that the optic nerve of the contralateral eye will become involved within days unless high-dose corticosteroid treatment is begun. The treatment is nearly 100% effective in preventing second eye involvement.

INFARCT

MORPHOLOGY:

First 4-12 hrs:

- No gross or microscopic change

By 12-18 hrs:

- Pink neurons
- Neuronal changes similar to global ischemia, but accompanied by PMNs

By 36-48 hrs:

- Swelling and softening of parenchyma and blurring of the demarcation between gray and white matter
- Hemorrhages

By 72 hrs:

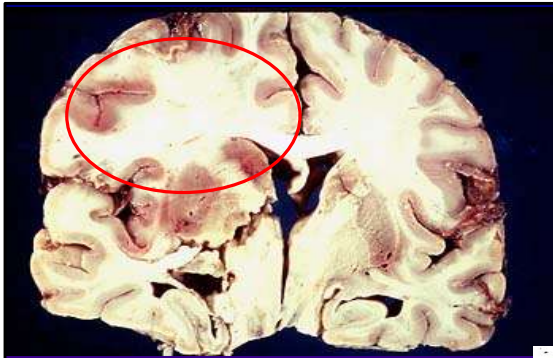
- Phagocytosis of the dead cells results in sharper demarcation of the infarct, peak macrophages at 7-14 days

By 1 month:

- Extensive phagocytosis results in further softening and liquefaction and development of irregular cavities

By 6 months:

- Complete liquefaction
- A thin rim of subpial parenchyma supplied by leptomeningeal arteries is preserved over a cortical infarct (helps to differentiate an old infarct from an old contusion).

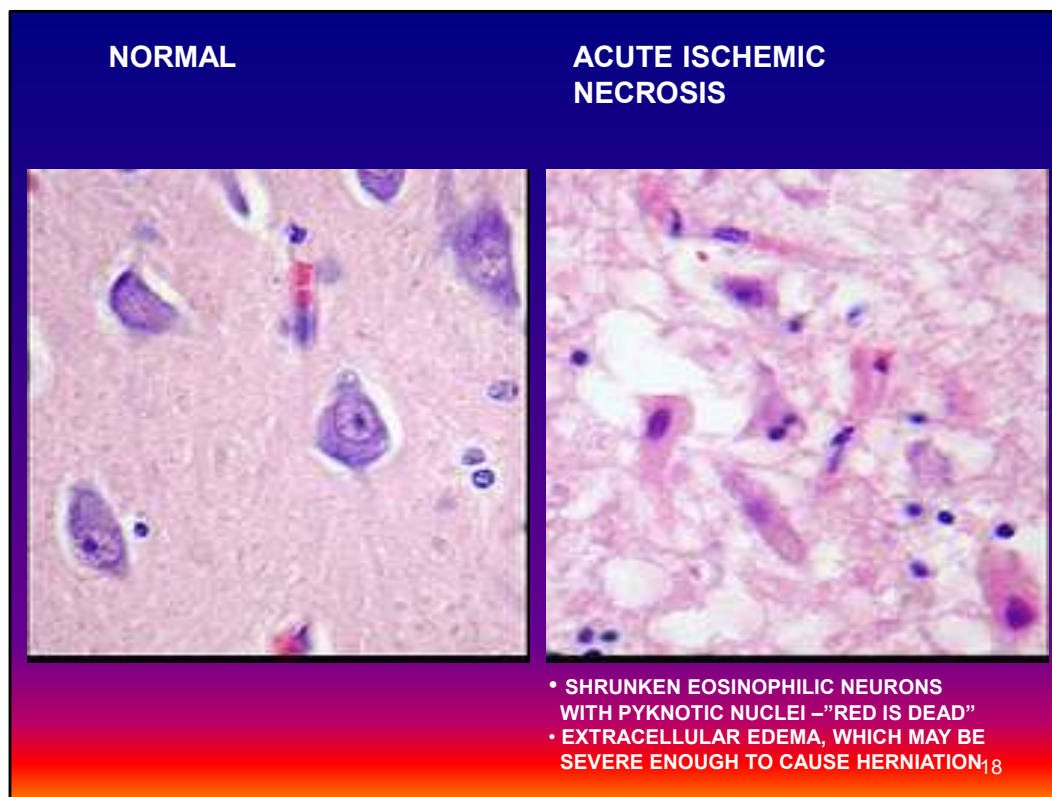


- **MASSIVE ACUTE INFARCT**
- CEREBRAL EDEMA CAUSED A MIDLINE SHIFT
- SECONDARY BLEEDING AT THE EDGES OF THE INFARCT

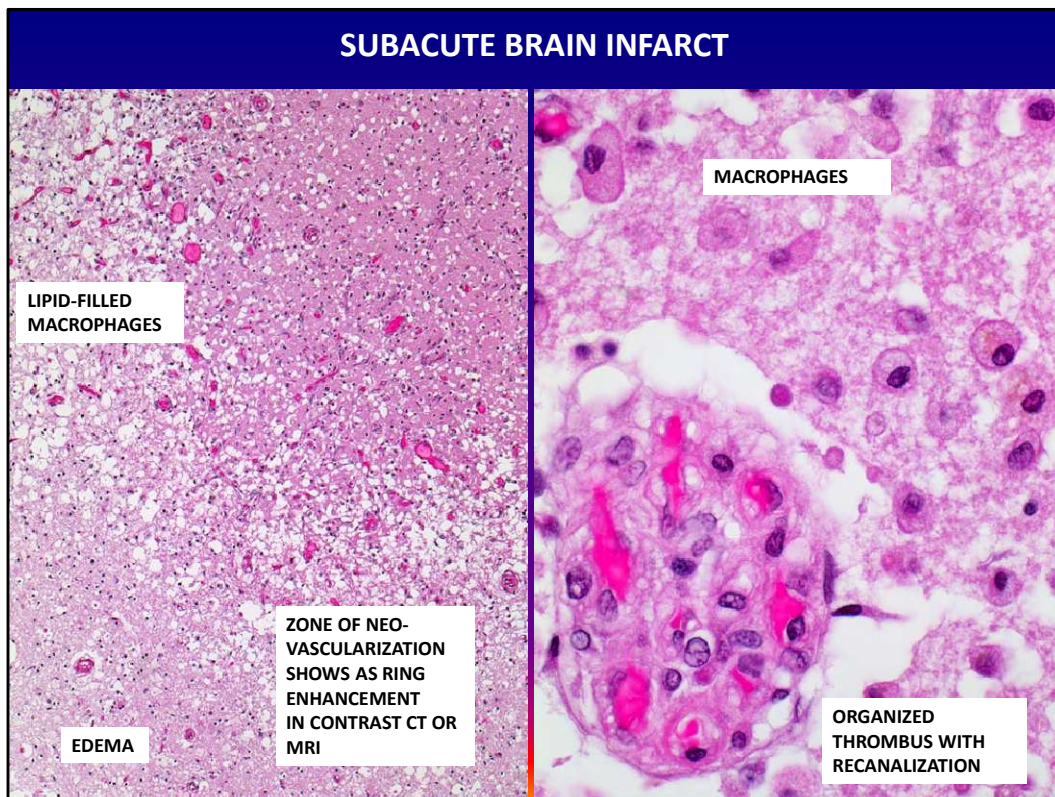
FRESH ANEMIC INFARCT

- THE NORMAL SHARP DEMARCATION BETWEEN THE GRAY AND WHITE MATTER IS LOST
- FOCAL HEMORRHAGES





Brain, normal (left) and acute necrosis (right) – High power

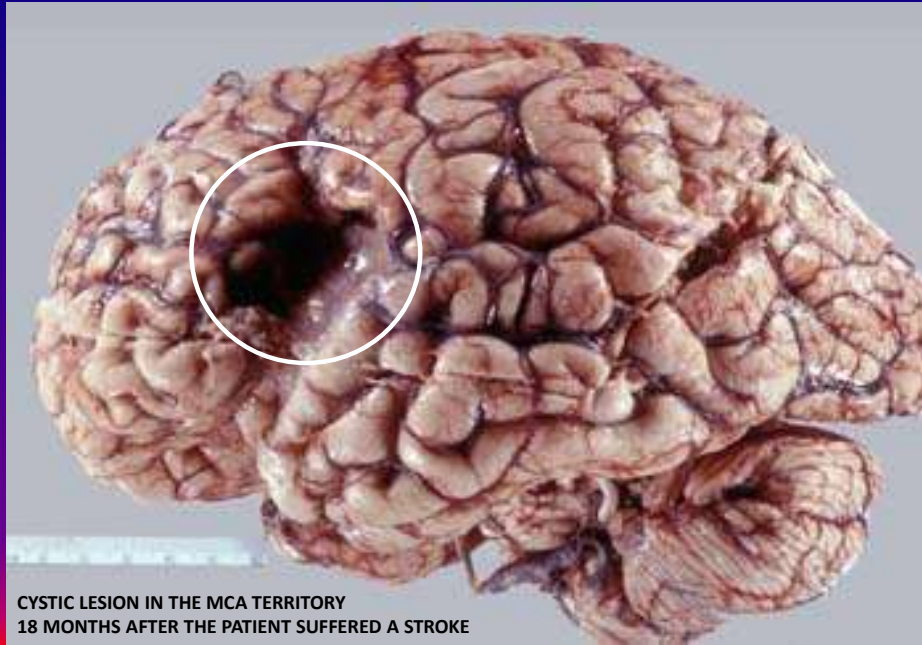


87 y o woman. Brain, parietal lobe. **Diagnosis:** Subacute brain infarct.

Left: Description: Edema and neovascularisation.

Right: Description: Organized thrombus with recanalization.

OLD ANEMIC INFARCT

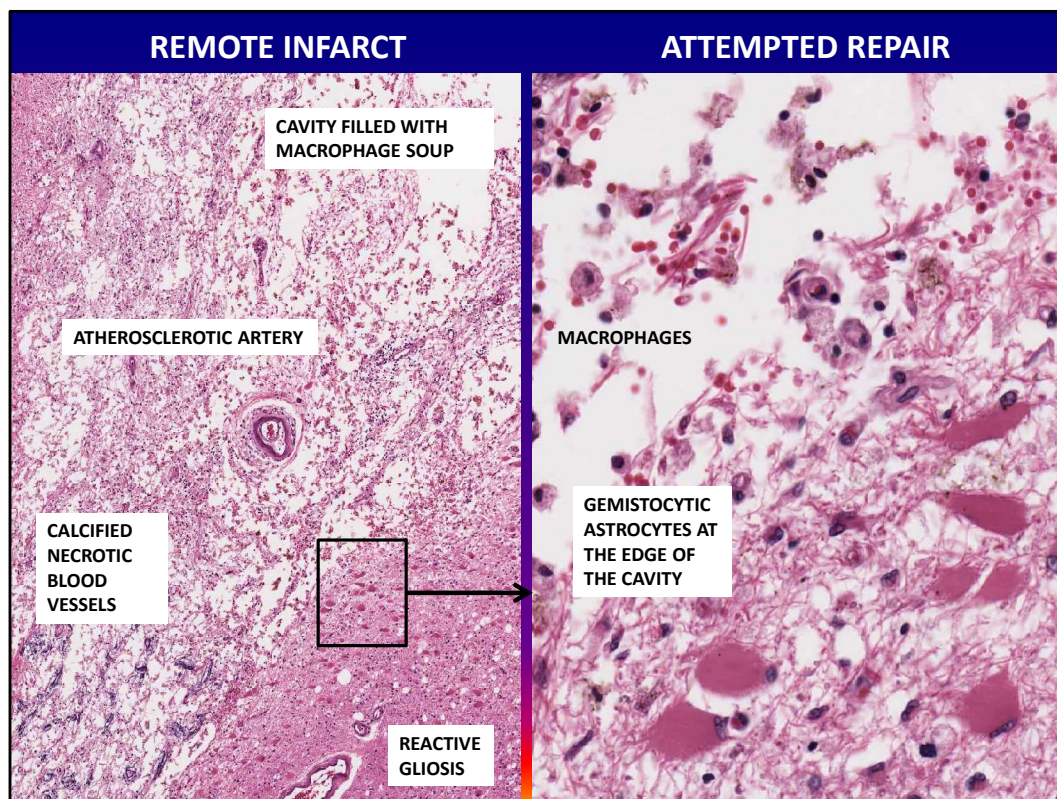


CYSTIC LESION IN THE MCA TERRITORY
18 MONTHS AFTER THE PATIENT SUFFERED A STROKE

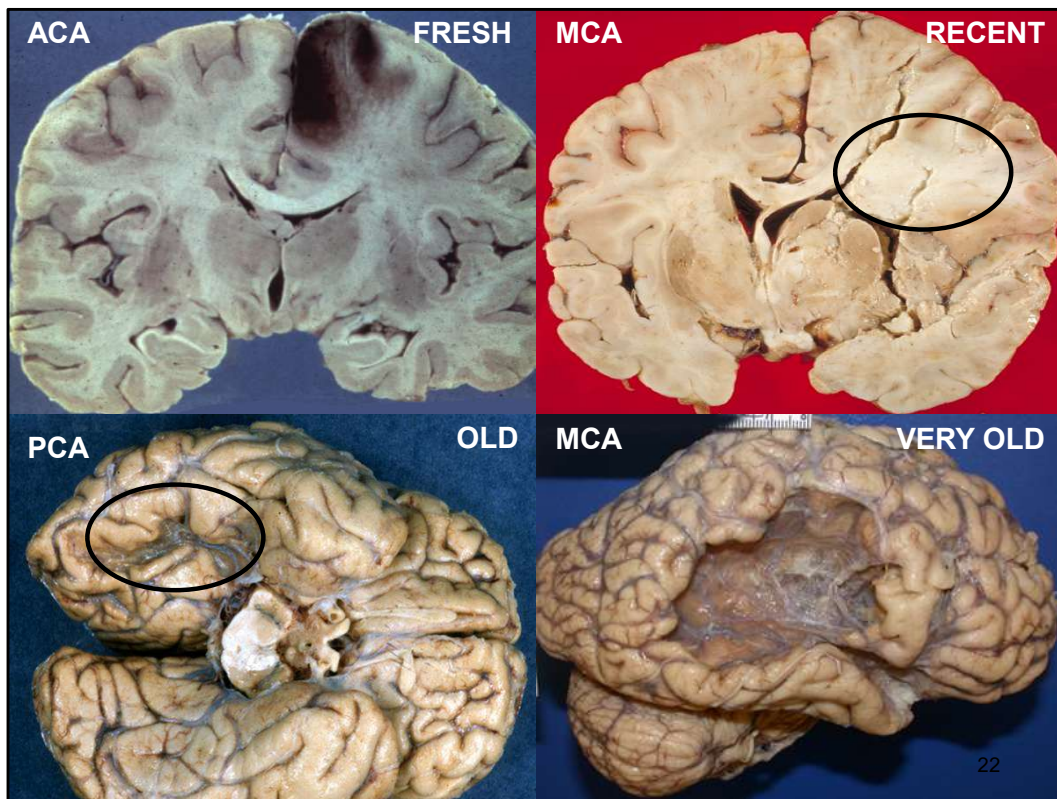
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Brain, gross, whole brain.

History: A 68-year-old right-handed woman awoke one morning aphasic and unable to move her right arm and leg. The patient died 18 months after the onset of the CNS symptoms.



Acc305-77 439.svs, Brain, HE, 5X



LACUNAR INFARCTS

- Small, slit-shaped, <1.5 cm in diameter
- Usually multiple
- Located at sites supplied by perforating branches:
 - Basal ganglia
 - Internal capsule
 - Thalamus
 - Pons
- Arteriolosclerosis (small vessel disease) associated with **chronic hypertension** is the most common cause
 - Endothelial injury and leakage of plasma proteins from vessels cause development of typical hyaline arteriolosclerosis
- Lacunar infarcts may be asymptomatic or cause focal neurologic deficits
- Multiple lacunar infarcts are associated with ischemic white matter degeneration and s.c. vascular dementia (Binswanger disease)
- Multiple lacunar infarcts in basal ganglia cause a syndrome of rigidity and abnormal gait resembling Parkinson disease



**MULTIPLE LACUNAR INFARCTS WITH WHITE MATTER ATROPHY RESULT IN
CLINICAL SUBCORTICAL VASCULAR DEMENTIA**

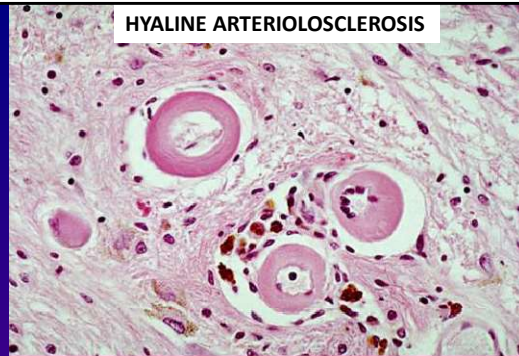


Brain– Gross, coronal section. Multiple lacunar infarcts are associated with ischemic white matter degeneration and dementia (Binswanger disease)

SMALL VESSEL DISEASE – OTHER EXAMPLES

Diabetic microangiopathy

- Hyaline arteriosclerosis
- Probably caused by glycation of proteins, which leads to diffuse basement membrane thickening and endothelial damage



HYALINE ARTERIOLOSCLEROSIS

Cerebral autosomal dominant arteriopathy with subcortical infarcts and ischemic leukoencephalopathy (CADASIL)

- Caused by mutations of the *notch3* gene in chromosome 19
- The problem is degeneration of vascular smooth muscle cells
- Migraine headache and TIAs beginning at ca 30 years of age with detectable tissue changes in white matter by imaging studies
- Recurrent strokes and dementia

Genetic angiopathy due to mutations of collagen 4A1

- Collagen 4A1 is a constituent of the vascular adventitia
- Causes porencephaly (CNS cysts and cavities) in infants – rare
- Causes lacunar infarcts and leukoencephalopathy in adults

INTRACRANIAL HEMORRHAGE

CLASSIFICATION AND PATHOGENESIS:

Based on the partition of the intracranial space into four anatomic compartments:

- Brain parenchyma
- Subarachnoid space
- Subdural space
- Epidural space

ETIOLOGY:

- Hypertension (Charcot-Bouchard microaneurysms)
- Berry aneurysm
- Trauma
- Vascular malformations
- Primary and metastatic neoplasms
- Bleeding associated with hematologic malignancies
- Coagulation disorders (anticoagulation therapy)
- Cerebral amyloid angiopathy (CAA)
- Vasculitis (polyarteritis nodosa, SLE, bacterial and fungal arteritis)

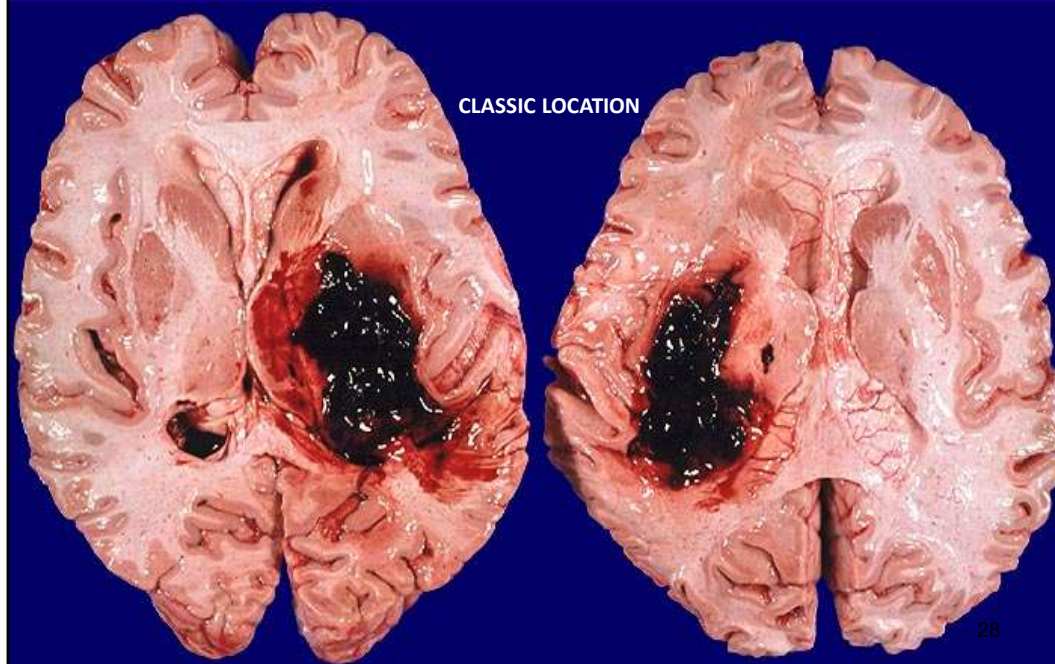
INTRACRANIAL HEMORRHAGE

INTRAPARENCHYMAL HEMORRHAGE:

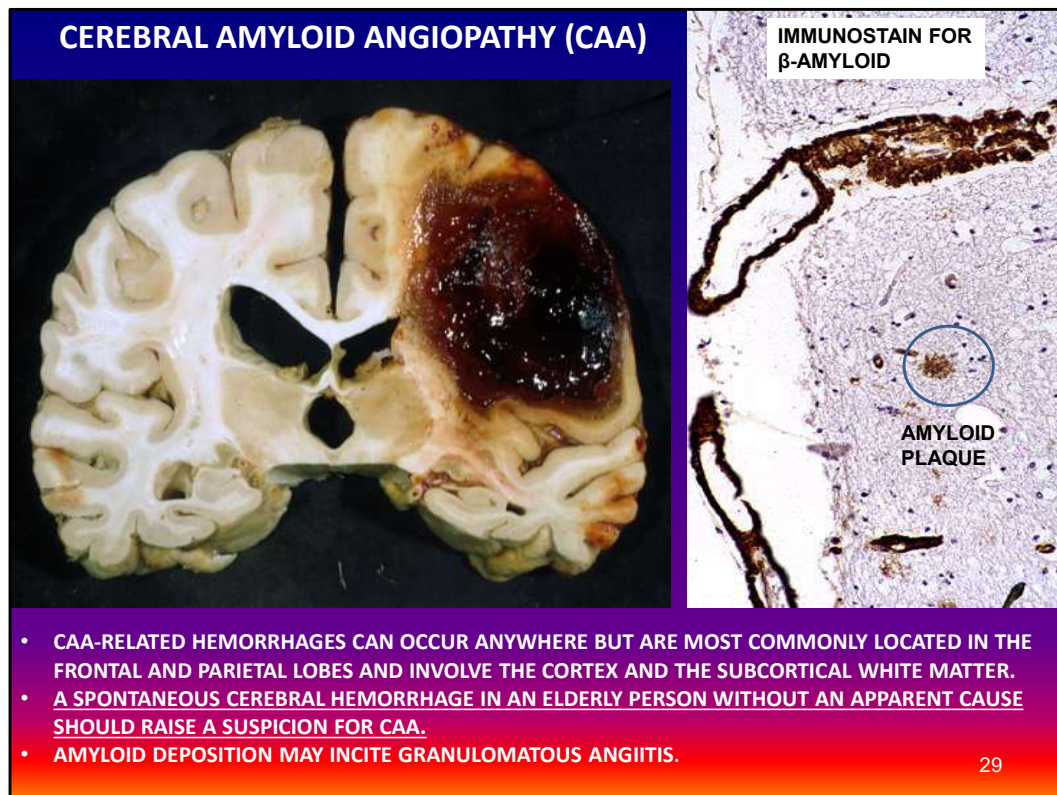
- Lesions are predominantly in three locations:
 - 65% in the basal ganglia and thalamus
 - 15% in the pons
 - 10% in the cerebellum
- Minute hemorrhages (petechiae) are less harmful than the obstruction of the associated small vessels
- Large hematomas act as space occupying lesions and have the potential to produce focal neurologic deficits and to cause lethal transtentorial herniation

FRESH MASSIVE HEMORRHAGE IN BASAL GANGLIA

RISK FACTOR: HYPERTENSION



Diagnosis: Fresh massive hemorrhage in basal ganglia (typical location) in a hypertensive patient. Hemorrhages involving the basal ganglia area (the putamen in particular) tend to be non-traumatic and caused by hypertension, which damages and weakens the small penetrating arteries. A mass effect with midline shift, often with secondary edema, may lead to herniation. Bouchard microaneurysms are the source of the hemorrhages.



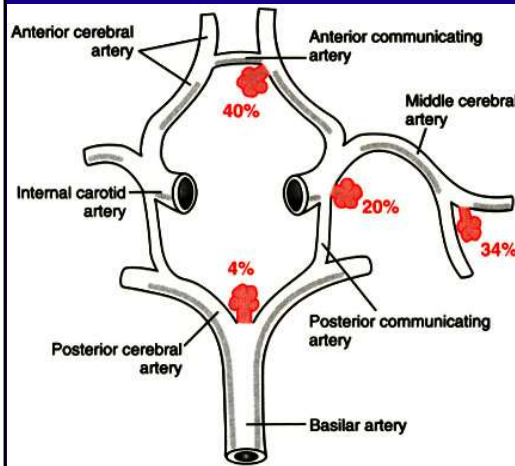
Left: Massive cerebral hemorrhage in a patient with cerebral amyloid angiopathy (CAA). **Right:** Immunostain for beta-amyloid.

- **Lobar distribution (in contrast with HTN-associated bleeds, which occur more often in thalamus and basal ganglia)**

SUBARACHNOID HEMORRHAGE

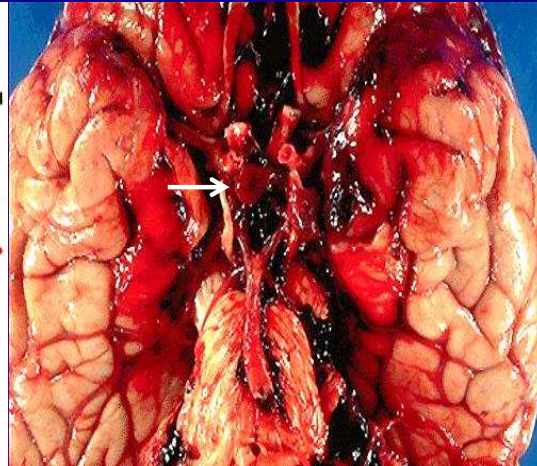
- Most common causes:
 - Rupture of a berry aneurysm
 - Trauma
- Less common cause:
 - Arteriovenous malformation
- Clinical signs:
 - Headache (increasing intracranial pressure)
 - Nuchal rigidity (meningeal irritation)
 - Alteration in mental status
- Complications:
 - Hydrocephalus resulting from organization of the blood in the CSF
 - Communicating:** CSF flows through the ventricles, but is not adequately absorbed into the bloodstream.
 - Non-communicating:** The aqueduct and/or the foramina are closed and the flow of CSF is impaired.

COMMON SITES OF BERRY ANEURYSMS



90% in the internal carotid artery territory
 <10% vertebral-basilar
 20-30% multiple

RUPTURED BERRY ANEURYSM



The subarachnoid hemorrhage from a ruptured aneurysm is more of an irritant producing vasospasm rather than a mass lesion.

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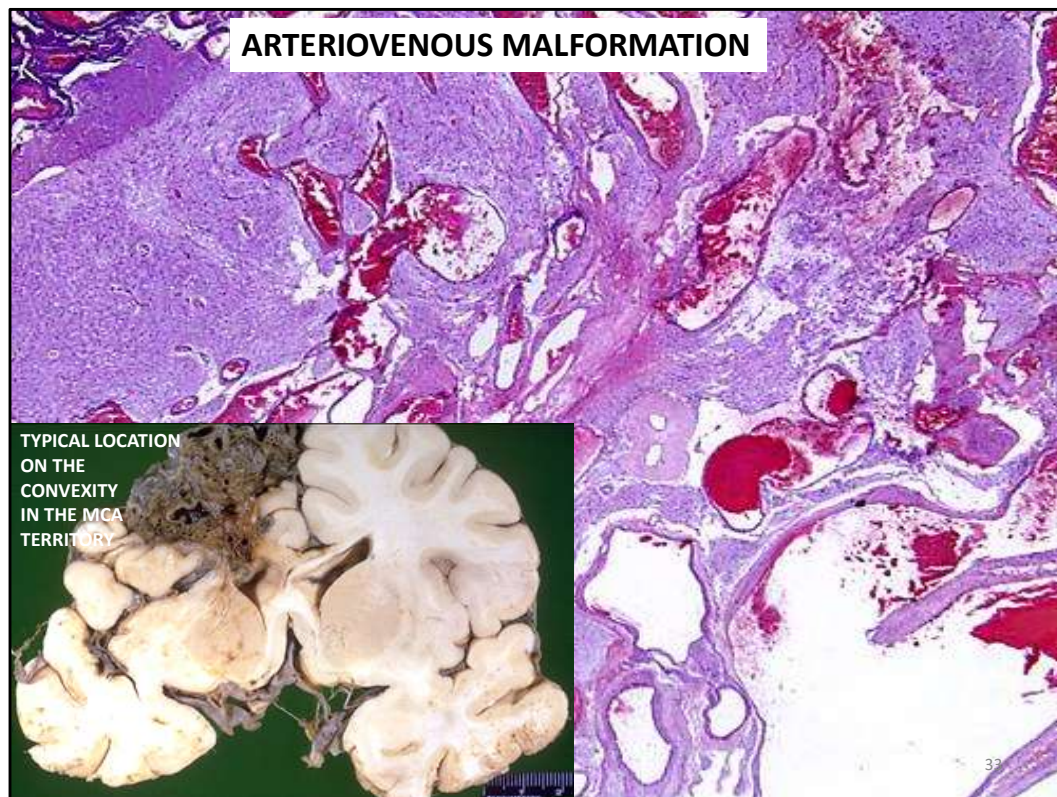
Common sites of saccular (berry) aneurysms in the circle of Willis. Such aneurysms are "congenital" in the sense that the defect in the arterial wall is present from birth, but the actual aneurysm takes years to develop, so that **rupture is most likely to occur in young to middle age adults.**

FUSIFORM ANEURYSM IN BASILAR ARTERY



- MIDDLE-AGED OR OLD PATIENTS
- DEVELOPS ON THE BASIS OF ATHEROSCLEROSIS
- IN-SITU THROMBOSIS CAUSING ANEMIC STROKE IN THE BASILAR ARTERY TERRITORY

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- Present at birth.
- Low pressure AV shunt
- Symptoms caused by mass effect and bleeding.
- Often seizures and meningeal irritation with slow leak
- Massive catastrophic bleeding kills the patient.
- Can be treated, if close to the surface, but deep lesions have poor prognosis, because surgery is difficult or impossible.
- Microscopically tortuous vascular channels with arterial and venous and mixed characteristics.

HEAD TRAUMA

- Penetrating or blunt; open or closed
- The magnitude and distribution of brain lesions depend on
 - the shape of the object causing the trauma,
 - the force of impact,
 - whether the head is in motion at the time of injury
- Severe brain damage can occur in the absence of external signs of head injury
- Severe lacerations and even skull fractures are not always associated with injury to the brain

CEREBRAL CONCUSSION

- **Mild form of diffuse axonal injury (DAI)** without permanent pathology
- Loss of consciousness is probably due to a functional disturbance of the reticular activating substance of the upper brainstem (the part of the CNS that is subjected to the highest twisting force during sagittal rotation of the hemispheres)

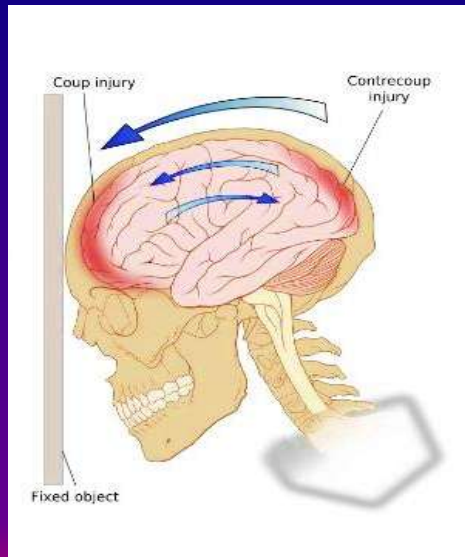
CEREBRAL CONTUSION

- Caused by rapid tissue displacement, disruption of vascular channels, and subsequent hemorrhage, tissue injury, and edema.
- The crests of the gyri are most susceptible to traumatic injury, because they are closest to the skull
- Contusions are common in regions, where the inside of the skull is irregular, e.g. the orbitofrontal regions and the temporal lobe tips.
- Contusions are wedge-shaped, with the widest aspect closest to the point of impact
- Blood seeps into the involved tissue across the cerebral cortex and into the white matter and subarachnoid spaces within hours.
- Function is impaired early, but morphologic changes (nuclear pyknosis, cytoplasmic eosinophilia, cellular disintegration) take about 24 hours to appear.
- Inflammatory response by PMNs is followed by macrophages.
- The superficial cortex is most severely affected in contrast with ischemic lesions, in which the superficial layer of cortex may be preserved.
- Coup injury occurs under the site of impact with an object.
- Contrecoup injury occurs on the side opposite the area of the impact.

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CEREBRAL CONTUSION - CONTRE COUP INJURY

ACUTE CONTRECOUP CONTUSION

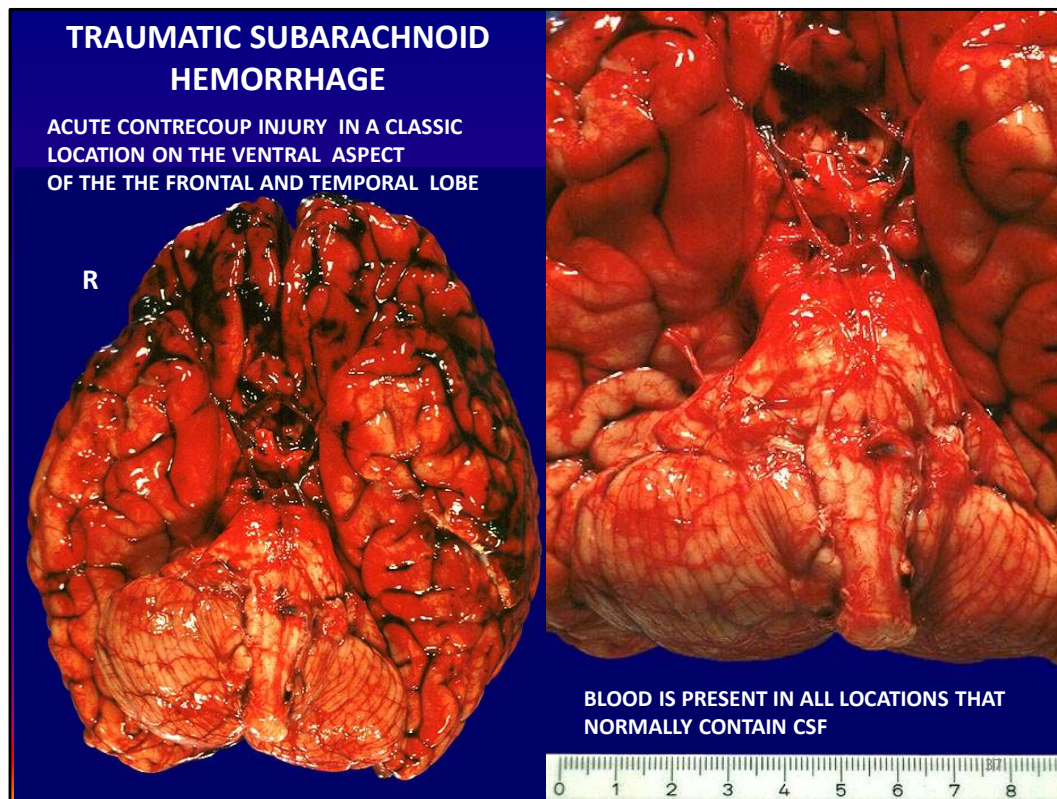


CONTUSIONS ARE COMMON IN AREAS, WHERE THE INSIDE OF THE SKULL IS IRREGULAR

- THE ORBITOFRONTAL REGION
- TIPS OF TEMPORAL LOBES

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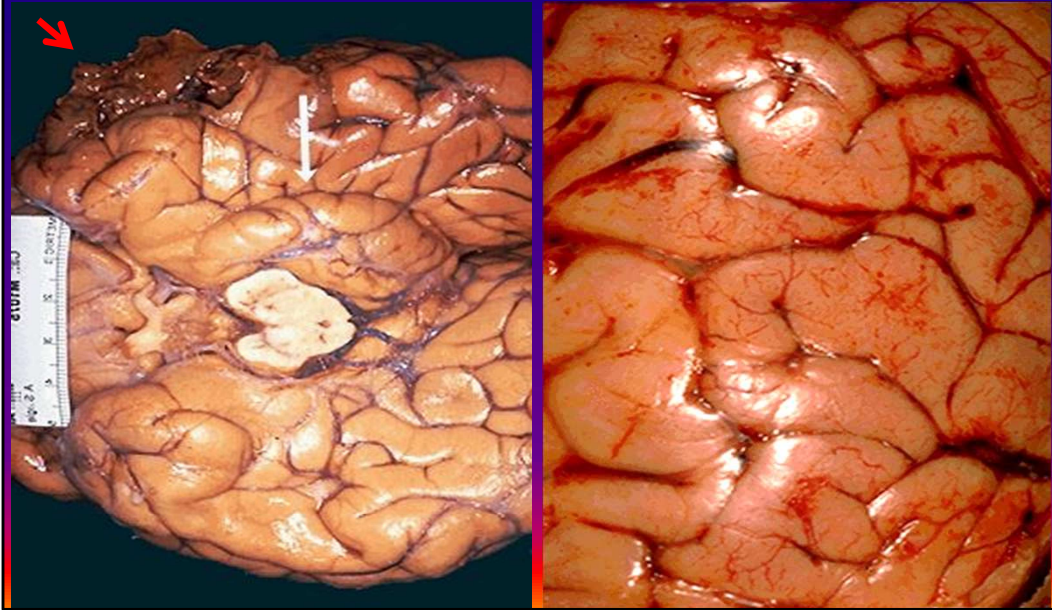
Brain. Gross, ventral view. The crests of the gyri are most susceptible to traumatic forces producing this injury pattern



43 y o man. Base of brain. **Diagnosis:** Fresh traumatic subarachnoid hemorrhage.
Coup contrecoup injury

LACERATION - PENETRATING HEAD TRAUMA

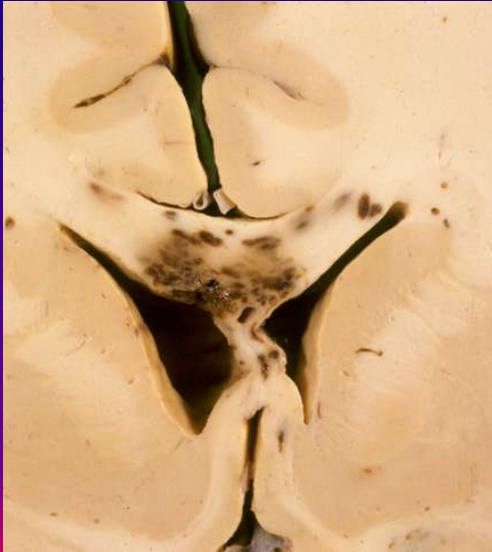
- LACERATION IN THE RIGHT HEMISPHERE
- BRAIN EDEMA WITH FLATTENING OF GYRI AND NARROWING OF SULCI
- UNCAL HERNIATION IN THE RIGHT HEMISPHERE



Brain – gross. Ventral aspect of brain and a view of the convexity. Acute brain swelling in the closed cranial cavity is a serious condition. There is a laceration in the right hemisphere. The brain is edematous with flattening and widening of gyri and narrowing of sulci. Swelling of the right cerebral hemisphere has produced a shift with herniation of the uncus of the hippocampus through the tentorium, leading to the groove seen at the white arrow.

DIFFUSE AXONAL INJURY (DAI)

DAI: HEMORRHAGES
IN CORPUS CALLOSUM



DAI: HEMORRHAGES IN BRAIN STEM

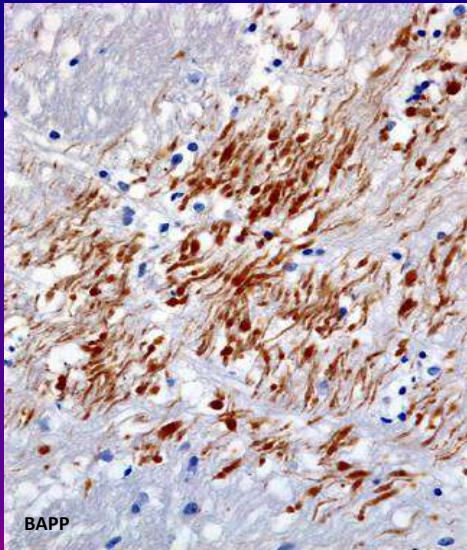


- CAUSED BY BLOWS TO AN UNSUPPORTED HEAD
- OCCURS MOST FREQUENTLY IN MOTOR VEHICLE ACCIDENTS

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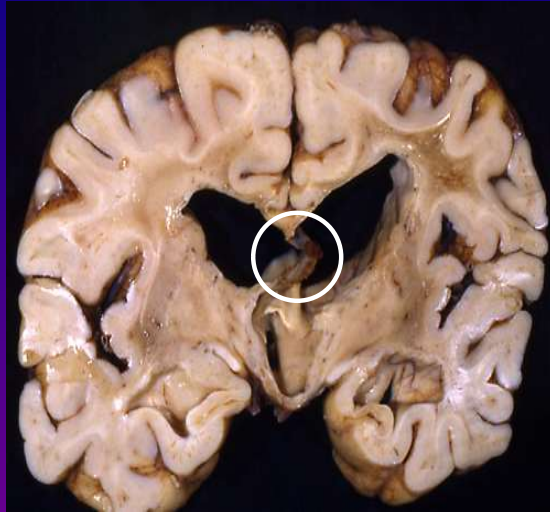
DIFFUSE AXONAL INJURY (DAI)

FRESH DAI: AXONAL SWELLING



- IMMUNOSTAIN FOR BETA-AMYLOID PRECURSOR PROTEIN (BAPP) SHOWS SWOLLEN AXONS
- BECOMES POSITIVE 2-3 HRS AFTER INJURY

OLD DAI:



CORPUS CALLOSUM ATROPHY AND SEVERE WHITE MATTER DEGENERATION

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Left: Axonal swellings can be detected with H&E and silver stains 15 hours after injury. Immunostains with antibodies to Beta Amyloid Precursor Protein (BAPP) can detect axonal lesions 2-3 hours after injury. BAPP is produced by neurons as a reaction to injury. **Right:** Brownish discoloration of white matter is due to old micro hemorrhages. **Glial fibrillary acid protein (GFAP) can be detected in blood 1-2 hours after injury and reaches its peak in 20 hours.**

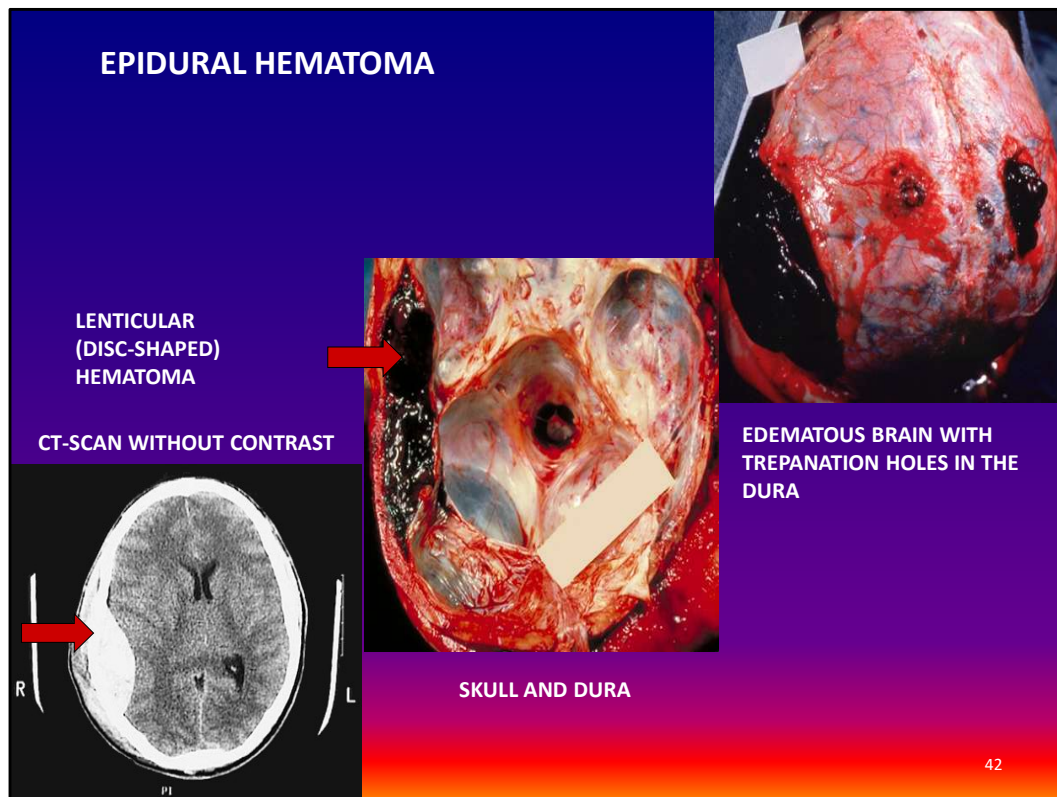
ACUTE TRAUMATIC ENCEPHALOPATHY MARKERS

- **Serum levels of biomarkers correlate with the degree of traumatic brain injury (TBI)**
- Used in the clinical settings to predict the severity of TBI.
- Used in Forensic pathology: there is a survival-time-dependent increase in astroglial GFAP and neuronal IL-6 expression in fatally injured human brain.
- **GFAP (Glial fibrillary acid protein)***
 - **Detectable in serum in <15 minutes after acute traumatic brain injury (ATBI)****
- **Ubiquitin C-terminal hydroxylase (UCH-L1)***
 - **Detectable in serum within an hour of injury; the level correlates with the severity of injury determined by imaging findings and the Glasgow Coma Score.**
- **IL-6**
 - Elevated levels immediately after the trauma are associated with a worse outcome.
 - Produced by reactive astrocytes, microglial cells and neurons
 - May contribute to disruptions of the BBB and the progression of cerebral edema.
 - Promotes neurogenesis and regeneration in animal models of TBI.
- **Neurofilament light chain (NFL)**
 - Released from damaged neurons (analogous to troponin release from myocardium).
- **Tau**
 - Significantly increased within hours of injury.

* The FDA authorized the use of a blood test for glial fibrillary acidic protein (GFAP) and ubiquitin carboxy-terminal hydrolase L1 (UCH-L1) in mild traumatic brain injury (mTBI) in 2018.

A cartridge that can measure GFAP and UCH-L1 in whole blood within minutes after a TBI used in a hand-held iSTAT device (which can also measure several other analytes in whole blood) was tested by the US Military and was cleared by FDA for diagnosing acute TBI on April 1, 2024.

The cartridge uses a sandwich enzyme-linked immunosorbent assay (ELISA) to measure the amount of GFAP in the sample.

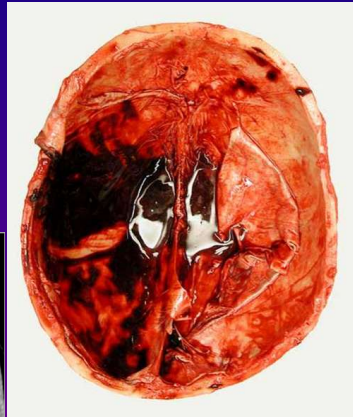
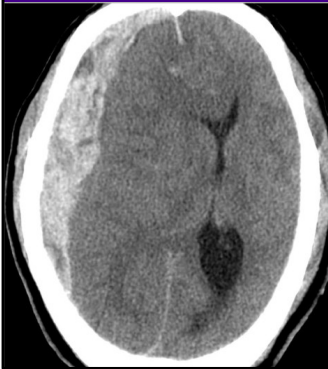


- CT- scan without contrast demonstrates a large epidural hematoma with right to left shift and ventricular narrowing.

ACUTE SUBDURAL HEMATOMA

NO-CONTRAST CT:

A LARGE ACUTE SUBDURAL
HEMATOMA ACTS AS A SPACE-
OCCUPYING LESION AND CAUSES A
MIDLINE SHIFT

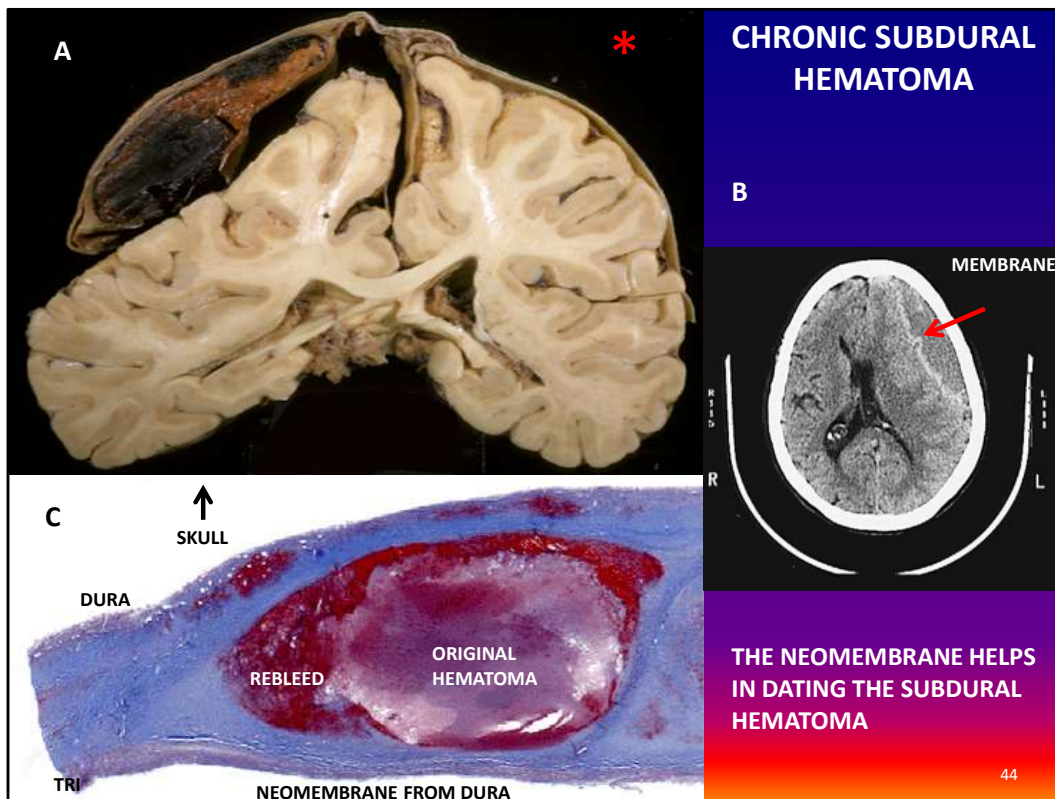


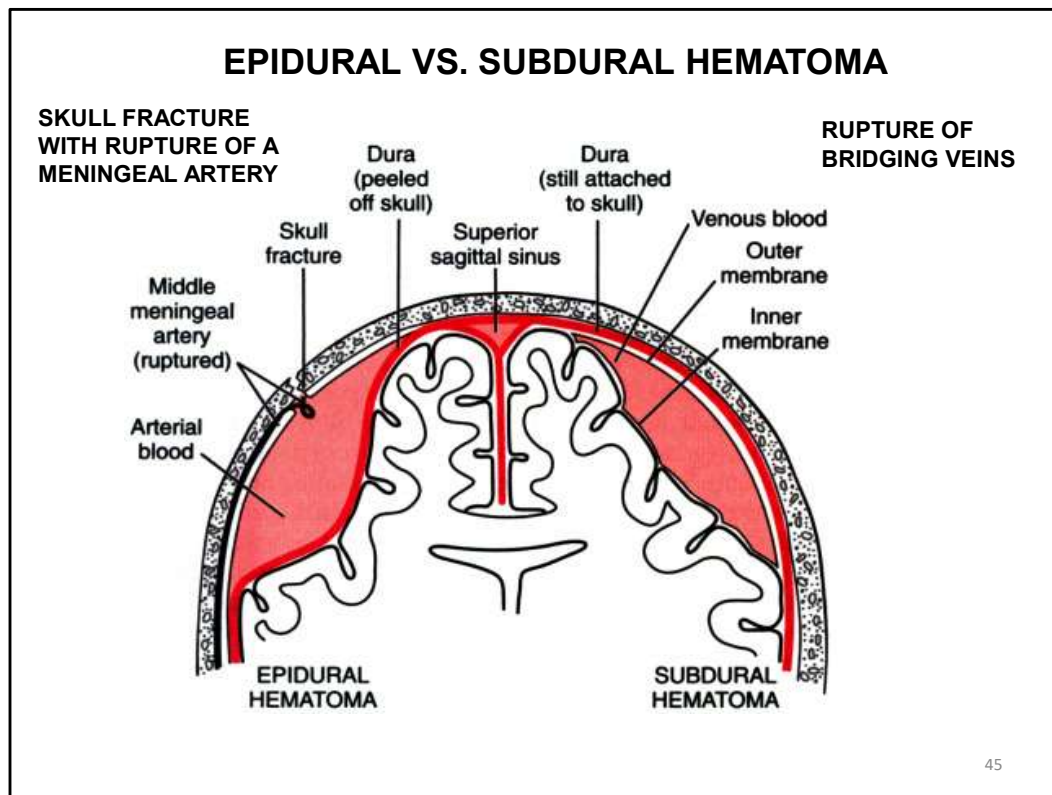
SKULL SIDE



BILATERAL SUBDURAL
HEMATOMAS, LARGER IN
THE LEFT HEMISPHERE

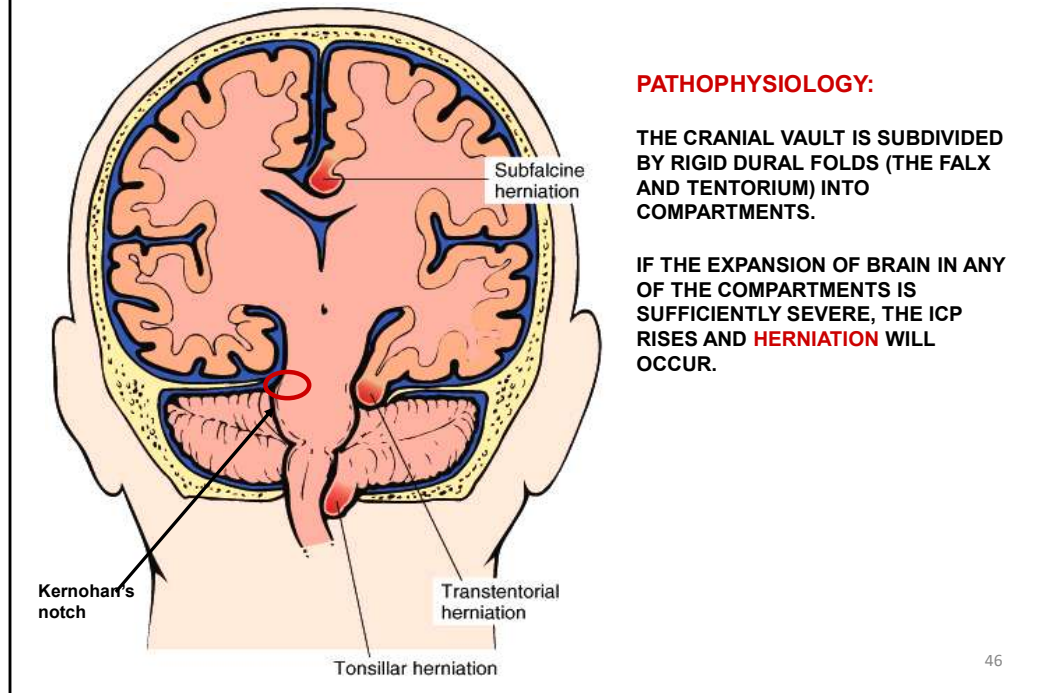
- Results from trauma that shears the bridging veins and/or small arteries that traverse the space between the arachnoid and the dura



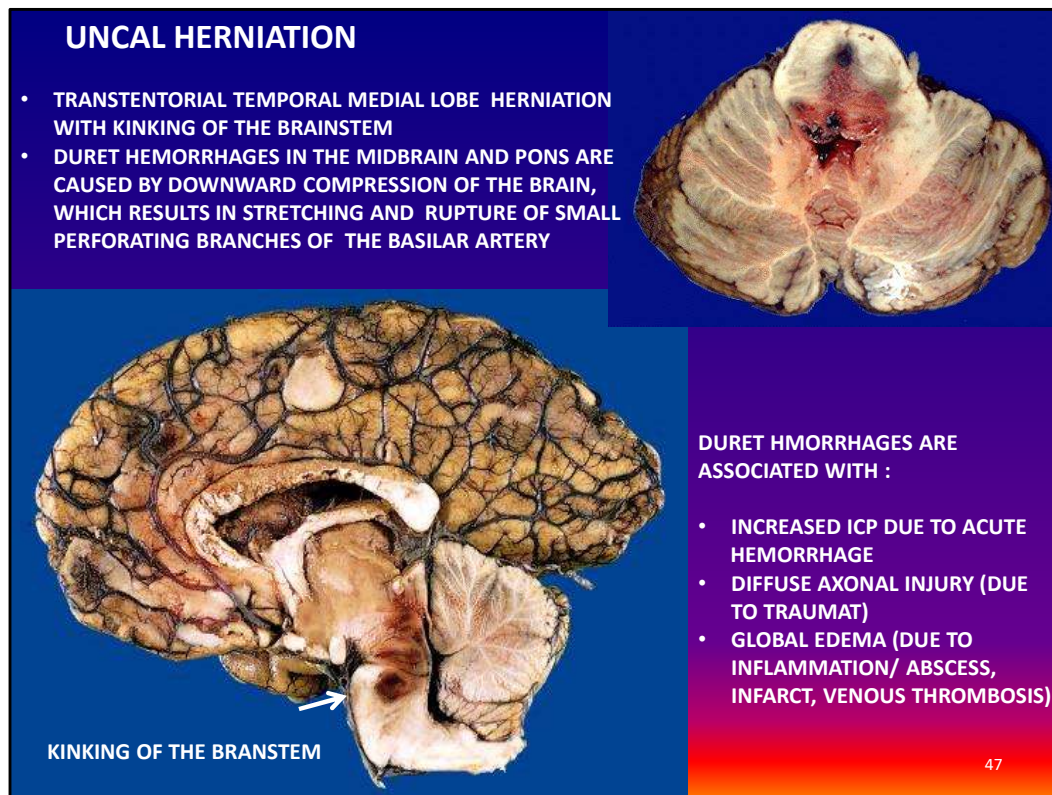


Robbins 7th Ed, Figure 28-11 Epidural hematoma (*left*) in which rupture of a meningeal artery, usually associated with a skull fracture, leads to accumulation of arterial blood between the dura and the skull. In a subdural hematoma (*right*), damage to bridging veins between the brain and the superior sagittal sinus leads to the accumulation of blood between the dura and the arachnoid.

MAJOR CNS HERNIATIONS



Robbins 10th ed, Figure 23-4. Major herniations of the brain: subfalcine, transtentorial, and tonsillar.



Brain – gross, mid-sagittal section and a cross section through midbrain and pons.

There is kinking of the brainstem due to herniation. Duret hemorrhages are present in the midbrain and pons. Duret hemorrhages are small linear areas of bleeding in the midbrain and upper pons of the brainstem. They are caused by downward displacement of the brainstem, which leads to stretching and rupture of small perforating branches of the basilar artery. The displacement is a result of uncal (temporal medial lobe) herniation due to increased intracranial pressure following acute hemorrhage, trauma (diffuse axonal "shearing" injury), tumor or inflammation/abscess. Infarction and venous thrombosis may cause global edema and elevated ICP leading to Duret hemorrhages. The diagnosis is made by CT or MRI. The prognosis is poor.

THE “SHAKEN BABY” SYNDROME

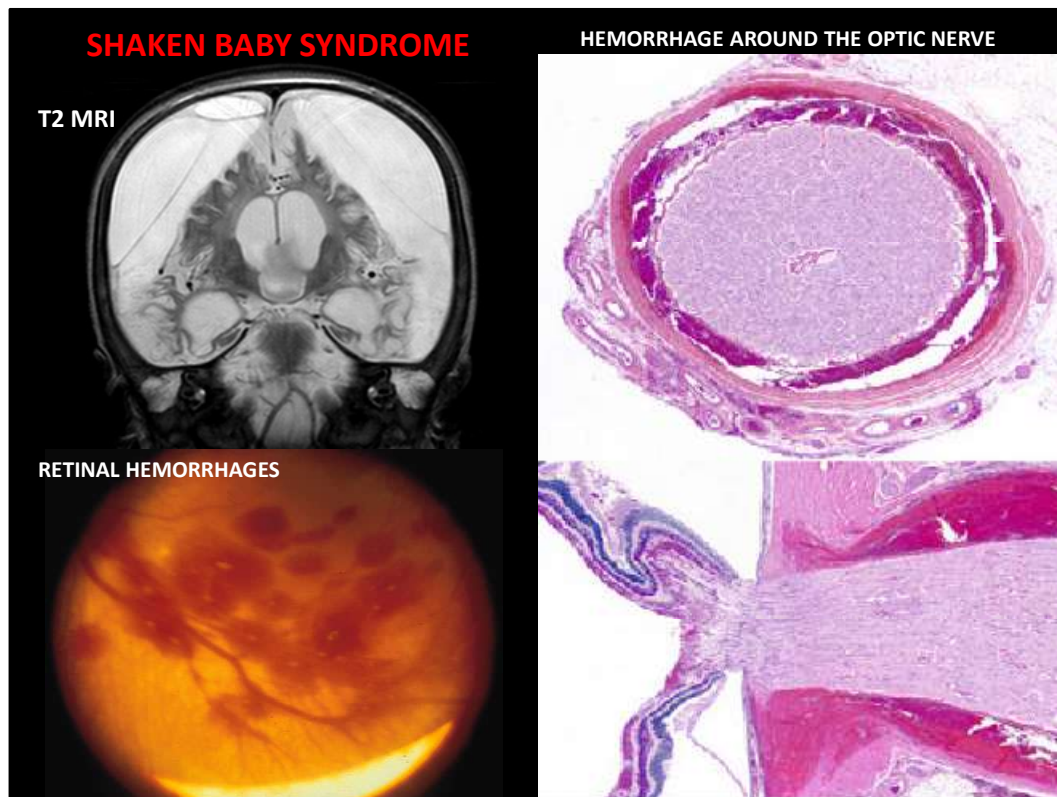
PREVALENCE:

- Child abuse is the second most common cause of death in children under one year of age, after the sudden infant death syndrome
- Estimated 50,000 cases of shaken baby syndrome in the United States each year
- Nearly 25% of infants with shaken baby syndrome die from the sustained brain injuries
- The victims range in age from just a few days to five years, with an average age of six to eight months
- Young men (early 20s) are the usual perpetrators

PATTERN OF INJURIES:

- Combination of subdural hematoma, subarachnoid hemorrhage, contusions, DAI, and retinal hemorrhage are often present
- Infants with these injuries are usually brought to the ER with a nonspecific clinical picture of hypotonia, listlessness, vomiting, irritability, and lethargy.
- **Diagnosis may be missed unless a skeletal survey and funduscopic examination are done.**

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T2 MRI. Severe brain atrophy and bilateral subdural hematomas. The subdural hematomas may have evolved from smaller ones that were present on admission. However, subdural hematomas will likely develop following such severe cerebral atrophy.

Retinal hemorrhages, often large and confluent, are an important component of the SBS and correlate with the severity of brain damage. They are probably caused by shaking of the vitreous body which pulls the retina and tears its delicate vessels. Documentation of retinal hemorrhage by funduscopy examination is important in the clinical evaluation of the SBS. **Retinal hemorrhages are uncommon in accidental brain injury** and are infrequently seen in patients with bleeding disorders, angiopathies, sepsis, and other nontraumatic settings. **The subdural hematomas and retinal hemorrhages are markers of the SBS.** Encephalopathy and death are caused mainly by **global HIE-the key pathology of the SBS-and increased intracranial pressure.** The back and forth motion of the head during shaking applies great stress to the craniocervical junction and the lower brainstem. In extreme cases, the spinal cord may separate from the brainstem. Lesser forces impair the function of vital centers in the brainstem causing central apnea. Cerebral edema which follows HIE raises intracranial pressure and makes HIE worse. Vascular tears from shaking cause microscopic or large parenchymal hemorrhages. Infarcts in an irregular geographic pattern are common. These lesions add to the neurological deficits and further raise the intracranial pressure. The subdural hematomas may add another increment. DAI may affect the cortex-white matter junction, the corpus callosum, internal capsule, or

the brainstem and contributes to the encephalopathy. Infants with the SBS usually also have rib, skull, and other fractures (often of varying ages), skin and soft tissue bruises, internal injuries, and other evidence of trauma.

THE END

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